higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL SYSTEMS AND CONSTRUCTION
NQF LEVEL 3

21 NOVEMBER 2019

This marking guideline consists of 7 pages.

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QUESTION 1

1.1 A certificate of compliance has a unique number✓ and it is obtainable from the chief inspector, or person appointed by the chief inspector,✓ in the form of an addendum, and issued by a registered person in respect of an electrical installation (or part of an electrical installation).✓ (3)

1.2 This expression means that circuits are electrically isolated✓ from other circuits in an electrical installation.✓ (2)

1.3 1.3.1 Indirect contact shock occurs when a live wire or part touches a piece of metal✓ and the body comes into contact with the live metal.✓

1.3.2
- Earthing and bonding
- Automatic disconnection of the supply
- Using doubly insulated equipment (Any 2 × 1)

(2 × 2) (4)

1.4

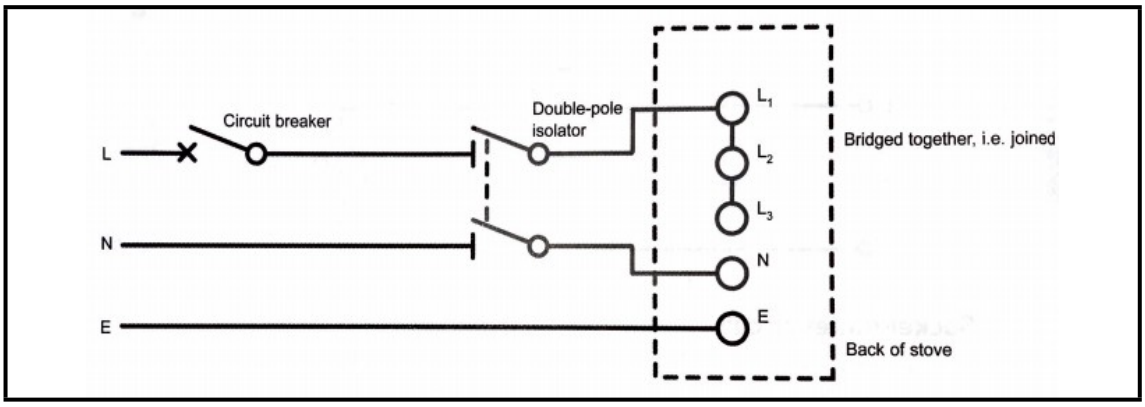


FIGURE 1

(5 × 1) (5)

1.5 Wire size = 1,5 mm^2
Circuit breaker rating = 10 A (2)

1.6 1.6.1 A coupling is used to extend the length of a conduit.

1.6.2 A male bush is used to protect conductors from protruding from a conduit.

(2 × 1) (2)

1.7 A draw tape or fish tape (1)

1.8
- Minimise the likelihood of damage to the insulation and subsequent failure of the distribution system.
- Ensure as far as possible the safety of personnel working in the area in which the cables are installed.
- Keep the overall cost of the installation to a minimum. (3 × 2) (6)

[25]

QUESTION 2

2.1
- There are no administration charges or hidden charges.
- The consumer can manage their own electricity bill by monitoring usage.

(Any 2 × 2) (4)

2.2
- Main switch
- Earth leakage relay
- Surge protection devices
- Circuit breakers of various sizes
- Load control relay
- Electronic timer
- Domestic contactor

(Any 4 × 1) (4)

2.3
- Accessible components are correctly selected.
- All protective devices are of the correct current rating.
- All conductors are of the correct current carrying capacity.
- Components have been correctly installed.
- Disconnecting devices are correctly located and all switch gear switches phase conductors only.
- Different circuits are electrically separated.
- Connection of conductors, earthing and bonding are mechanically strong and electrically continuous.
- Circuits, fuses, switches, terminals, earth leakages, circuit breakers and distribution boards are correctly and permanently marked.
- Safety lighting and emergency lighting and signs are functioning correctly.
- In the case of a new installation or alterations to an existing installation, the new, added or altered part complies with SANS 10142-1.
- In the case of an existing installation, the installation complies with the general safety principles of SANS 10142-1.
- Where an alternative supply is installed, it complies with the requirements of section 7.12 of SANS 10142-1.

(Any 6 × 1) (6)

2.4
- Poor probe contact
- Weak crocodile clips
- Faulty (intermittent) leads
- Using leads other than those supplied with an instrument.
- Incompetent user

(Any 2 × 1) (2)

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2.5

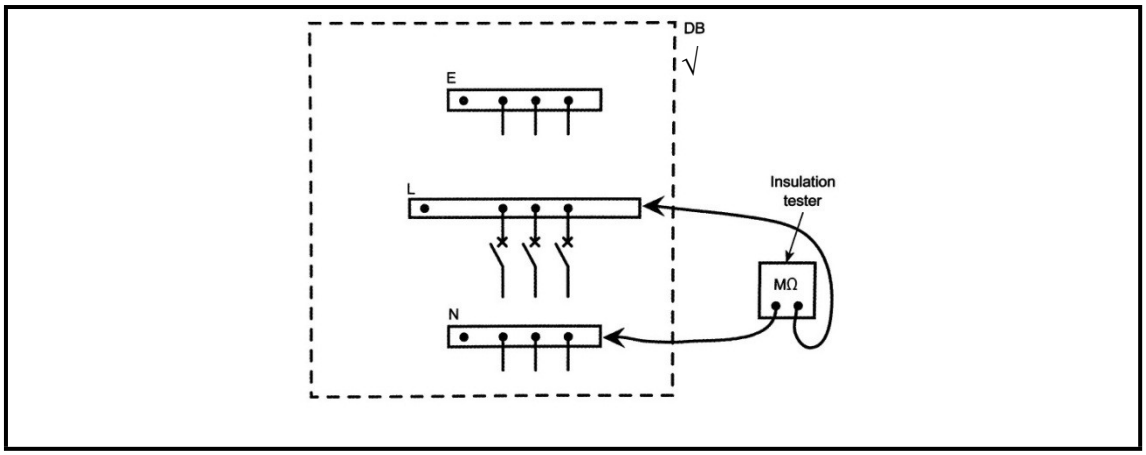


FIGURE 2

(5 × 1) (5)

2.6 Disconnect the installation✓ and notify the supplier.✓ (2)

2.7 As per regulation 6.2.7.1.1 of SANS 10142-1: 2012:

$V_{drop} = V_{ph} \times 5\%$✓
$= 11,5$ V✓ (2)

[25]

QUESTION 3

3.1
- There is no power.
- There is an open circuit or loose connection.
- The element is faulty. (Any 2 × 1) (2)

3.2
- Electrocution
- Electric shock
- Burns
- Falls (Any 2 × 1) (2)

3.3
- Visual defects
- Smell
- Timing of occurrences
- Loss of functionality
- Intermittency
- Sound
- Heat/temperature (Any 4 × 1) (4)

3.4 3.4.1
- Cause: The conductor resistance is too high.
- Corrective action: Replace the conductors with a minimum size of 6 mm^2.

3.4.2
- Cause: There is an open circuit in the geyser element.
- Corrective action: Replace the geyser element.

(2 × 2) (4)

3.5
- High-risk tasks during which incidents have or may have the potential to result in serious injury
- When any part of a hazard assessment or risk analysis shows a high or sufficient risk to the worker
- When incidents occur frequently
- New tasks or tasks where equipment has been added, replaced or modernised
- When two or more workers must perform specific tasks simultaneously

(Any 3 × 1) (3)
[15]

QUESTION 4

4.1 A
4.2 B
4.3 C
4.4 D
4.5 A
4.6 B
4.7 C
4.8 D
4.9 A
4.10 B

(10 × 1) **[10]**

QUESTION 5

5.1 Tungsten (1)

5.2
- Single-coil filaments
- Coiled-coil filaments (2)

5.3

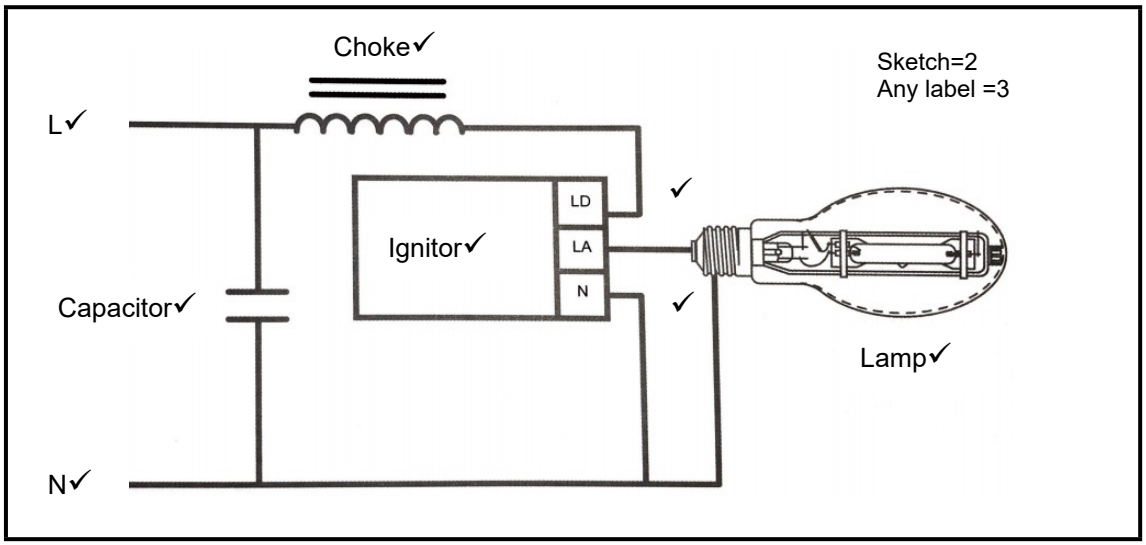


FIGURE 3 (5)

5.4
5.4.1 Pins
5.4.2 Base
5.4.3 Lead-in wires

(3 × 1) (3)

5.5
- Drawing offices
- Classrooms
- Halls (Any 2 × 1) (2)

5.6
- Increases losses in power distribution systems and results in increased energy costs.
- Larger cables and switchgear are necessary to supply a given load.
- Larger currents give rise to greater copper losses in transmission lines and transformers.
- Larger currents give rise to greater volt drops in cables.
- Larger cables may be required on the consumer's side of the electrical installation to carry the larger currents demanded by a load operating at a low-power factor.
- Larger cables required lead to higher installation costs.
- The consumer pays a higher electricity bill. (Any 2 × 1) (2)

[15]

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QUESTION 6

6.1
- Missing or loose rungs
- Damaged or worn non-slip feet
- Loose screws, rivets, bolts and nuts
- Loose or faulty locks, hinges, and other metal parts
- Cracks and exposed fibreglass
- Sharp edges on rails and rungs
- Corrosion, rust and excessive wear on treads
- Twisted or distorted rails
- Missing identification labels
- Mud, grease, oil or wet paint on the rungs, tiles, steps or platform

(Any 2 × 1) (2)

6.2
6.2.1 False
6.2.2 True
6.2.3 True
6.2.4 False
6.2.5 True
6.2.6 True
6.2.7 False
6.2.8 False

(8 × 1) (8)
[10]

TOTAL: 100

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